

Quantifying the Impact of the 2023 Nova Kakhovka Dam Collapse on Rapeseed Crop Yield in Vasylivskiy, Ukraine Using Sentinel-2 Data

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Abstract

On June 3, 2023, the Nova Kakhovka Dam in Kherson, Ukraine collapsed. The immediate effect was felt downstream where towns and cropland was flooded. The long-term consequences were felt more broadly, as miles of canal systems fed by the Kakhovka reservoir upstream from the dam ran dry. In particular, this impacted the agricultural sector in SE Ukraine where the canals brought vital water for the irrigation of crops.

This paper uses remote sensing to examine the long-term impact of the dam collapse on the Vasylivskiy district's rapeseed crop yield. The paper seeks to test whether the long-term impact of the dam collapse can be detected using remote sensing, and to quantify the impact in terms of crop yield lost in the years 2023, 2024, and 2025.

Since February 2022, the region has been an active conflict zone, making the collection of ground data difficult. The methods used in this paper demonstrate how remote sensing can be used to monitor disaster impact in regions of the world where ground truth data is difficult to collect.

1. Introduction

1.1 Nova Kakhovka Dam Collapse

Prior to the start of the Ukraine-Russia war in 2022, Ukraine was a bread basket of the world, exporting 10% of the world's wheat, corn and barley as well as agricultural products such as fertilizer (Becker-Reshef and Mitkish, 2024). Even after the war, its agricultural sector has remained resilient and been a vital food supply for the region (FAO, n.d.). On June 3, 2023, this vital center of food production came under immense pressure when the Nova Kakhovka dam collapsed.

Studies quantified how flooding from the dam collapse impacted agriculture immediately following the event and also predicted what the longer term effects might be. The London Stock Exchange Group estimated that 10,000 hectares of farmland were flooded downstream on the Ukrainian-controlled side of the river when the dam collapsed (Zhou, 2023).

The longer-term impacts were of even more concern. In the Kherson region alone, 95% of irrigation systems relied on water from the Kakhovka reservoir (Zhou, 2023). A UN report estimated that, in total, 306,500 hectares of crop land would be affected by this loss of irrigation, having to rely on ground water abstraction or rainfall instead and potentially reducing crop yield

by 70% (United Nations, 2023). Reuters put the affected acreage estimate at 500,000 hectares (Reuters, 2023).

However, there were few studies that quantified the longer-term impact after the fact. One study does quantify land abandonment rates in Ukraine more broadly since the start of the war, but it does not translate this into crop yield (Hartman et al., 2025). Additionally, Kussul and team at the National Technical University of Ukraine in Kyiv continues to collect in situ data and maintain a crop cover map of Ukraine (Kussul et al., 2025), but their work has not focused specifically on the dam collapse.

This paper aims to develop a method of quantifying the dam collapse's longer-term effect on agriculture by examining change in crop cover area and crop yield. The method is developed on a single crop type in a single district of Ukraine as a model that could then be applied to multiple crop types and to the larger affected region in Ukraine. The method could also prove useful for assessing the impact of events in conflict zones on crops more broadly, as it relies only minimally on *in situ* data.

1.2 Study Area

The study area for this project is a district just south of the Kakhovka Reservoir (Figure 1). It is an area that relies on canals fed by the reservoir (BBC News, 2023). The time period of the study is from 2019- 2025, which includes the dam collapse year of 2023 as well as the four years prior and the two years after.



Figure 1. Map showing the project study area of Vasylivskyi, Ukraine, with water masked out.

1.3 Rapeseed Crop Cover

Ukraine's six major crops are wheat, sunflowers, maize, soybeans, barley and rapeseed (USDA FAS, n.d.). This study focuses on rapeseed because its distinctive yellow flowers make it easier to distinguish from other crops using remote sensing (Han et al., 2025). Rapeseed is planted in the fall and harvested the following spring (USDA FAS, n.d.). Its flowers come into full bloom before the plant goes to senescence and is harvested. The NDYI index has been shown to be effective at distinguishing rapeseed during its full flower stage (Han et al., 2025).

2. Data

Sentinel-2 Level-2A (SR) 2019-2025. European Union Copernicus Satellite (ESA, n.d.). Sentinel-2 bands 2-4 have a 10m resolution, and bands 5-7 and 11-12 have a 20m resolution. The revisit interval is 5 days at the equator and 2-3 days at mid latitudes. The high temporal resolution lends itself well to studying phenological patterns. The spatial resolution is also useful for analyzing crop cover, because at 10m, the average field in Ukraine contains multiple Sentinel-2 pixels.

Eurocrop Map 2022 Covering EU-27 and Ukraine. European Commission Joint Research Center (Kwon et al., 2024). The crop map is developed from Sentinel-1, Sentinel-2 and *in situ* data and shows 19 crop types. In 2022, the model developed on EU data was applied to Ukraine and validated based on Kussul and team's reference data (Kussul et al., 2025). The team's Ukraine cropland map is available for view at <https://ukraine-cropmaps.com/> and the data itself is available upon request.

3. Methodology

3.1 Crop Classification

I constructed binary classifications of rapeseed crop cover and other landcover with a Random Forest model in Google Earth Engine (GEE). Using the European Commission Joint Research Center's 2022 map of Ukraine crop cover as ground reference data, I generated a binary raster of 2022 rapeseed and non-rapeseed pixels in QGIS. I then imported this map into GEE and generated 400 random sample points—200 rapeseed field points and 200 non-rapeseed points. These were the points I used to sample a mean image of the study area. I built the mean image from a collection of Sentinel-2 images ranging from dates 2022-05-01 to 2022-05-31. I chose the month of May to sample from because rapeseed in Ukraine typically flowers at this time of year, making it spectrally distinct from other crop types. The collection was filtered for cloud cover less than 20% and had a cloud mask applied. I then calculated the Normalized Difference Vegetation Index (NDVI) and the Normalized Difference Yellowness Index (NDYI) indices and added these as bands. NDVI compares the near infrared and red bands and effectively distinguishes vegetative land cover from non-vegetative land cover (Tucker, 1979). I ran the following equation for NDVI in GEE:

$$NDVI = \frac{NIR - Red}{NIR + Red}$$

I also added a band for the NDYI index which shows yellowness (Sulik and Long, 2016). The equation I implemented for this band in GEE was:

$$NDYI = \frac{Green - Blue}{Green + Blue}$$

In total, 14 images remained as part of the collection after filtering for dates and cloud cover, and these were used to create the mean May image. The mean May image included the following bands: B1, B2, B3, B4, B5, B6, B7, B8, B8A, B9, B11, B12, NDVI and NDYI.

Once the mean May image had been sampled, I used 60% of the sample data to train the model, reserving 40% for model validation. I used the Statistical Machine Intelligence & Learning Engine (Smile) Random Forest model (Li, 2017) which is directly available as a function in GEE. I kept all the parameters as default and used 50 decision trees to generate the model.

After training the model, I created mean May images for the years 2019-2021 and 2023-2025. Some of these years did not have sufficient cloud-free pixels in May to make a complete image, so in addition to the collection creation method described above for May 2022, I also patched the missing pixels with a mean June image with the same cloud filter and cloud mask settings.

I then applied the Random Forest model to the mean May image of each year, producing a rapeseed, non-rapeseed classification for each year. I added as properties the classified crop cover image's year, the total number of rapeseed pixels, and the total area of rapeseed as hectares so as to easily see the total area of rapeseed crop cover for each year.

I evaluated the model performance on 2022 validation data (details in the results section) and since there was no validation data available for other years, I performed visual checks on each year, comparing the classified map to the year's RGB mean image when rapeseed flowering could clearly be seen. This process revealed that the model was successfully capturing rapeseed crop cover, but that it was including some non-rapeseed pixels as rapeseed. This issue was most apparent in the post-dam collapse years of 2024 and 2025, where the model was categorizing large swaths of the newly contaminated water pixels as rapeseed. In order to solve this problem, I adjusted my study area boundary to not include the Dnieper River, effectively masking out those pixels from the study. (All seven crop cover maps are available in Appendix C.)

3.2 NDVI and NDYI timeseries

Using the annual rapeseed crop cover maps, I generated an NDVI and NDYI timeseries for rapeseed for each year. These were generated for each year individually and created as charts and CSV's in GEE.

3.3 Annual Rapeseed Crop Yield Estimate

In order to estimate crop yield using remote sensing imagery, I relied on a study by Lukas and team which explored the relationship between various spectral indices and rapeseed crop yield in a controlled experiment. The team took spectral data measurements at the BCCH 65 plant stage. The BBCH is a system for measuring phenology. It breaks a plant's lifecycle into 10 principal stages (Ministry of Agricultural, Food and Forestry Policies of Italy, 2001). For a rapeseed plant, BCCH 65 is the stage of full flowering. The crop cover of Lukas' team's study area was broken into a grid. The spectral data collected at the BCCH 65 point in time by an unmanned aerial vehicles (UAV) camera was compared to the yield of that grid. The NDYI index was found to have a strong linear relationship with crop yield, with an R^2 value of 0.9021 (Lukas et al., 2022).

In situ observations of BCCH 65 have been shown to correspond with peak yellowness of a Sentinel-2-derived NDYI timeseries (d'Andrimont et al., 2020) I generated an NDYI timeseries for each year to find the timing of peak yellowness. I attempted to smooth the timeseries using a moving average smoother of 30 days and one of 15 days, however the smoother lowered the values of peak NDYI without providing useful interpolation, so instead I used the raw values to find peak NDYI.

I recorded the peak NDYI value from each year's NDYI timeseries and then used this value in the d'Andrimont linear regression to estimate a per hectare yield.

$$y = 0.0778x + 2.9395$$

Where y is the annual yield per hectare and x is the NDYI value at the time of full flowering (BCCH 65).

I then multiplied each year's per-hectare yield by the crop area for that year.

3.4 Yield Loss Estimate

From the four full years of Sentinel-2 data that were available for the study area prior to the dam collapse, I calculated the range of crop area and crop yield. I then compared the years 2023, 2024 and 2025 individually to these ranges to see whether or not area and yield fell within the range of the four pre-dam collapse years. If they did not, I calculated how far outside the range they fell.

4. Results

4.1 Crop classification model

The Random Forest model used to create each year's crop cover map was trained on 2022 data and tested on 2022 data. There were 400 sample points in total and 60% (237 samples) were used for training, while 40% (163 samples) were reserved for validation. The GEE algorithm did not apportion the same proportion of each category to training and validation

though. There were more rapeseed samples in the training data than other landcover samples, and more other landcover samples than rapeseed in the validation data.

The model had an Overall Accuracy of 98.2% and a Kappa value of 0.963 (Table 1A), indicating that the model was highly accurate in distinguishing rapeseed crop cover from other land cover types. The confusion matrix showed a slightly higher user's accuracy for rapeseed, but a slightly higher producer's accuracy for other landcover types (Table 1B). For this use case, a high user's accuracy was important since it indicated that the pixels evaluated for NDYI were indeed likely to be rapeseed pixels. The model accuracy was not tested for other years as 2022 was the only year with ground reference data available. However, visual checks showed that once water was masked out the model performed well at detecting rapeseed pixels.

Table 1. Random Forest model evaluation metrics

A.

Kappa Value	0.963
Overall Accuracy	98.2%

B.

	Other Landcover	Rapeseed	UA
Other Landcover	88	2	98%
Rapessed	1	72	99%
PA	99%	97%	

4.2 Crop cover area for each year

The maximum extent of rapeseed crop cover during the four pre-dam collapse years of 2019-2022 was 60,999.66 hectares in 2019, and the minimum was 30,919.89 hectares in 2022. (Figure 2) Within the four years, the crop cover extent seemed to roughly follow an every other year pattern, where every other year the rapeseed crop cover was closer to 30,000 hectares and the other year closer to 60,000. However, there are not enough years to establish a pattern. (All seven crop cover maps are available in Appendix C.)

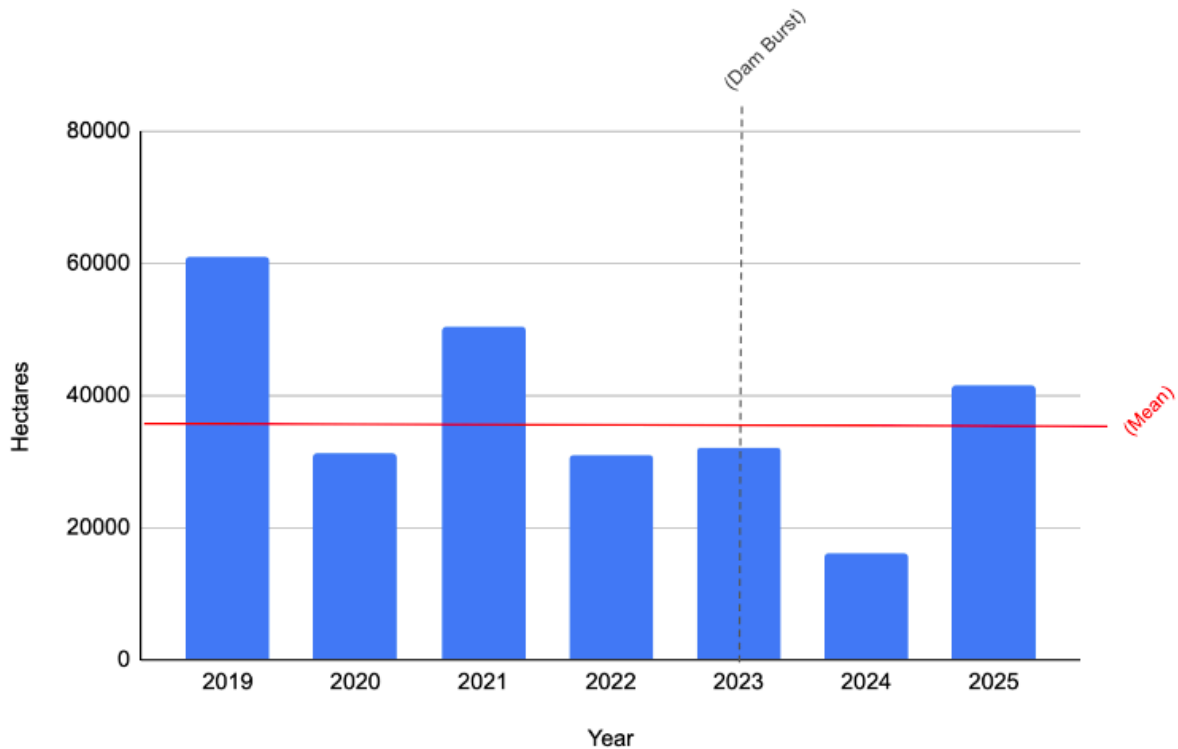


Figure 2. 2024 crop cover area was 16224.84, well below the 7-year mean of 37,630.67.

The dam collapsed in June 2023, while the crop cover extent map was calculated in the previous month of May, so one would not expect to see an effect on crop area in 2023. This does seem to be the case. The crop area in 2023 was 32,213.35 hectares, well within normal range from the four years previous.

In 2024, the dam collapse's effect on crop cover would seem to be significant. Total area in hectares for the year drops to 16,224.84 (Figure 2). However, in the following year we already see a quick recovery with crop cover back within pre-dam collapse norms, with a total area of 41,461.37 hectares.

When analyzing rapeseed crop cover area alone, without considering crop health or yield, these numbers seem to show that the district of Vasylivskyi's rapeseed crop was more negatively impacted by the immediate consequences of the dam collapse than by the long-term consequences of the lack of water to feed the reservoir and canals.

4.3 NDVI and NDYI timeseries

The NDVI and NDYI timeseries show the typical phenological pattern of the rapeseed crop. Rapeseed in Ukraine is planted in September or October and harvested the following year in July or August (USDA FAS, n.d.). Looking at NDVI, green up begins around March and peaks, on average, in May, after which the plant goes into senescence. NDYI follows a similar pattern,

but tends to peak after NDVI (Table 2). (The full set of NDVI and NDYI timeseries charts for all seven years are available in the appendix. Fig A andB)

Table 2. Value and Day or Year of Peak NDVI and Peak NDYI

Year	Peak NDVI	Peak NDVI DOY	Peak NDYI	Peak NDYI DOY
2019	0.736	147	0.377	162
2020	0.69	91	0.341	121
2021	0.704	146	0.362	176
2022	0.746	116	0.38	171
2023	0.716	156	0.362	126
2024	0.726	125	0.325	120
2025	0.766	117	0.326	125
Mean	0.73	128	0.35	143

The timeseries reveal information about how the 2023 dam collapse impacted rapeseed crop. The impact is more apparent in the NDYI timeseries than in NDVI (Figure 3). In the three years after the dam collapse NDVI remains above 0.70 at its peak. NDYI however, shows a drop to approximately point 0.32 in 2023 and 2024, and is not fully recovered to normal in 2025. Additionally, 2023 and 2024 are the only years in the study period where peak NDYI occurs before peak NDVI (Table 2), indicating that the crop is not growing as normal.

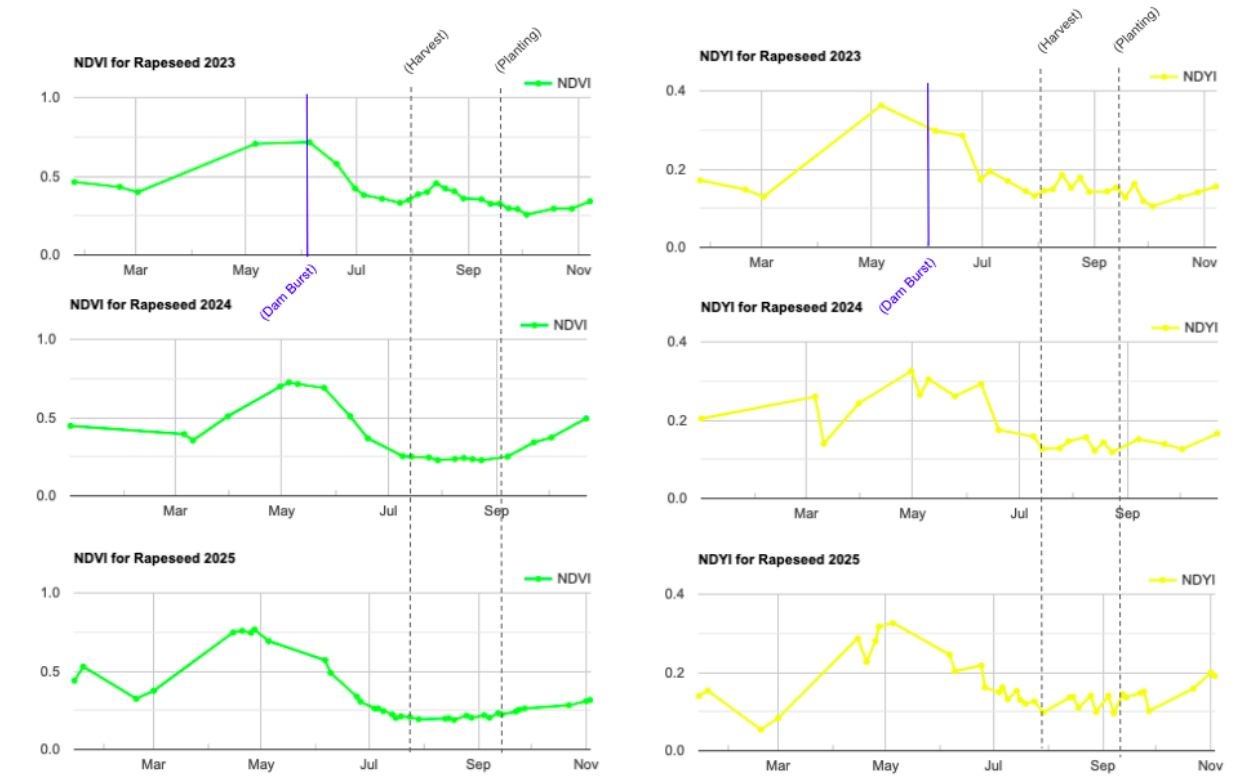


Figure 3. NDVI and NDYI timeseries for the three years impacted by the Kakhovka Dam collapse. NDYI is lower than normal, and occurs earlier than normal in 2023 and 2024.

4.4 Yield loss estimate

Combining the information from crop area and the NDYI timeseries in the manner detailed in the methodology section shows the patterns of crop yield over time and the impact of the dam collapse on crop yield. Crop yield ranges from a maximum of 181,097.66 tons and a minimum of 91,803.13 tons during the four years preceding the dam collapse (Table 3).

In 2023, yield does not appear to be affected. However, as mentioned above, this study's metric of measuring yield relies on NDYI at peak bloom, which in 2023 occurred in early May, prior to the dam collapse. The metric, therefore, does not capture whether crops were damaged or successfully harvested.

Table 3. Annual hectares of rapeseed crop cover and annual yield in tons.

Year	Hectares	Peak NDYI	Tons/Hectare	Yield in Tons
2019	60999.66	0.377	2.9688	181,097.66
2020	31209.88	0.341	2.9660	92,569.43
2021	50385.68	0.362	2.9677	149,527.75
2022	30919.89	0.38	2.9691	91,803.13
2023	32213.35	0.362	2.9677	95,598.39
2024	16224.84	0.325	2.9648	48,103.16
2025	41461.37	0.326	2.9649	122,927.27
Mean	37,630.67	0.35	2.97	111,660.97

In 2024, yield was certainly affected. Total yield for 2024 was 48,103.16 tons. This is 43,699.97 less than the lowest yield year and 132,994.5 less than the highest yield year. In 2025, yield was back within a normal range at 48,103.16 tons.

In summary, the metric was not able to reveal the dam collapse's effect on crop yield in 2023. In 2024 the yield loss was between 43,699.97 and 132,994.5 tons, and in 2025 the yield had returned to a normal range.

5. Conclusion and discussion

5.1 Errors and uncertainties

The errors and uncertainties in the project can be split into two categories: those related to the phenomenon studied and those related to the data and methods. This project was in part chosen in order to explore methods for obtaining information in areas where on the ground data is difficult to come by. The fact remains though that evaluating information in a conflict zone is very challenging. There are three reasons for this. First, ground truth data can be difficult to systematically collect and make available to the public. Second, it is difficult to establish a norm to compare to. The years preceding the dam collapse are dynamic and the events happening in Ukraine at that time may have already changed the norm of agricultural production in Vasylivskyi. Third, the many outside the norm factors at play that make cause and effect difficult to determine. Is the result due to dam collapse or is it due to other conflict-related events that may also have occurred at the same time? These questions must be kept in mind when evaluating the results.

There are also errors and uncertainties due to the data and methods. First, though Sentinel-2 data was important for its high spectral resolution and critical for its high temporal resolution in order to establish a phenological cycle, the dataset only goes back to 2017, and only has robust coverage of the study area starting in 2019. This meant a limited amount of data was available to quantify a pre-event norm. The "normal", pre-dam collapse years of 2019-2022 include the

years of a global pandemic in 2020 and an invasion by Russia in 2022. The short timeframe of data availability compounded the on-the-ground challenges of establishing a reliable norm.

Another issue of data availability was that validation data was only available for 2022. This introduces more uncertainty as to how the model performed in other years. The model performance could not be quantified during those years. This uncertainty was mitigated by visual checks of the models compared to the RGB images at full flower. These visual checks led to an important correction of masking out water pixels, which improved the model's performance in later post-dam collapse years.

Finally, another set of uncertainties was introduced in the calculation of yield based on the linear regression of its relationship with NDVI. The uncertainties with peak NDVI come from the temporal resolution of Sentinel-2 data. Though NDVI would not be lower than recorded, it is possible that Sentinel-2 missed a higher peak just previous to the one recorded. Two potential issues also exist with applying the linear regression to this project. First, the linear regression from the referenced study is based on NDVI calculated from data obtained by UAV, not Sentinel-2. Secondly the NDVI values of this project fall outside the referenced study's tested values. The mean peak NDVI value for this project was 0.35, but the lowest tested value in the referenced study was 0.47.

5.2 Future work

Despite uncertainties, this project lays out a methodology that could be a strong starting point for future work. There are two possible routes for further development of this project. The first is obtaining additional data to build out the strength of the analysis. In order to establish a more robust pre-event norm, I could attempt to use Landsat data and see if the temporal frequency is sufficient to detect a phenological pattern and whether that pattern corresponds with the Sentinel-2 pattern already obtained. If they are a match, Landsat could provide a more reliable normal range for crop area and also help determine whether there is a pattern or cycle for the amount of rapeseed planted from year to year. Another way to establish these patterns could be to obtain annual crop yield tables from Ukrainian government sources. If these tables are aggregated at the district level, they could be used to both corroborate this study and help establish the pre-event norm. Finally, the data from the Kussul team in Kyiv (Kussul et al., 2025) would be very useful in validating the model for other years besides 2022.

A second route of project development would be to build on the methods already begun here. First of all, I would like to add more data to the model for training by either feeding the model monthly mean values of each month of the year, or feeding it phenology statistics for the crop cover. Integrating a full year's worth of data would help in distinguishing crop types. The model already performs quite well without this additional data, but the data might be useful if I were to introduce other crop types into the analysis.

With this methodology established, it would not be difficult to expand the analysis to include yield impact estimates for all four major Ukrainian crops of rapeseed, wheat, maize and

sunflowers. This would only require minor changes to the methodology. However, I would have to find suitable studies that estimate crop yield based on spectral data for each crop type.

Finally, once worked out at the level of a single district, this project could be expanded to analyze the dam collapse impact on a larger region, including the two to three Ukrainian provinces that have been most impacted by the event.

5.3 Conclusions

This study quantified the impact of the 2023 dam collapse on rapeseed production in Vasylivskiy, Ukraine. It was inconclusive when it came to the impact on 2023 crop yield, but revealed some anomalies in the timing of peak NDVI that year. The study quantified a crop yield loss of between 43,699.97 and 132,994.5 tons in 2024, and showed a return to a normal range of crop yield in 2025. This indicates that despite the importance of canal systems fed by the Kakhovka Reservoir, rapeseed crop yield in Vasylivskiy was able to quickly recover from the event, and the most impact was felt in the immediate wake of the collapse.

This paper also provides a methodology for monitoring crop yield using remote sensing in parts of the world where, for reasons of conflict, ground truth data is difficult to obtain. The methods demonstrate that despite the uncertainties introduced in such a situation, critical monitoring data can still be obtained.

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Appendix

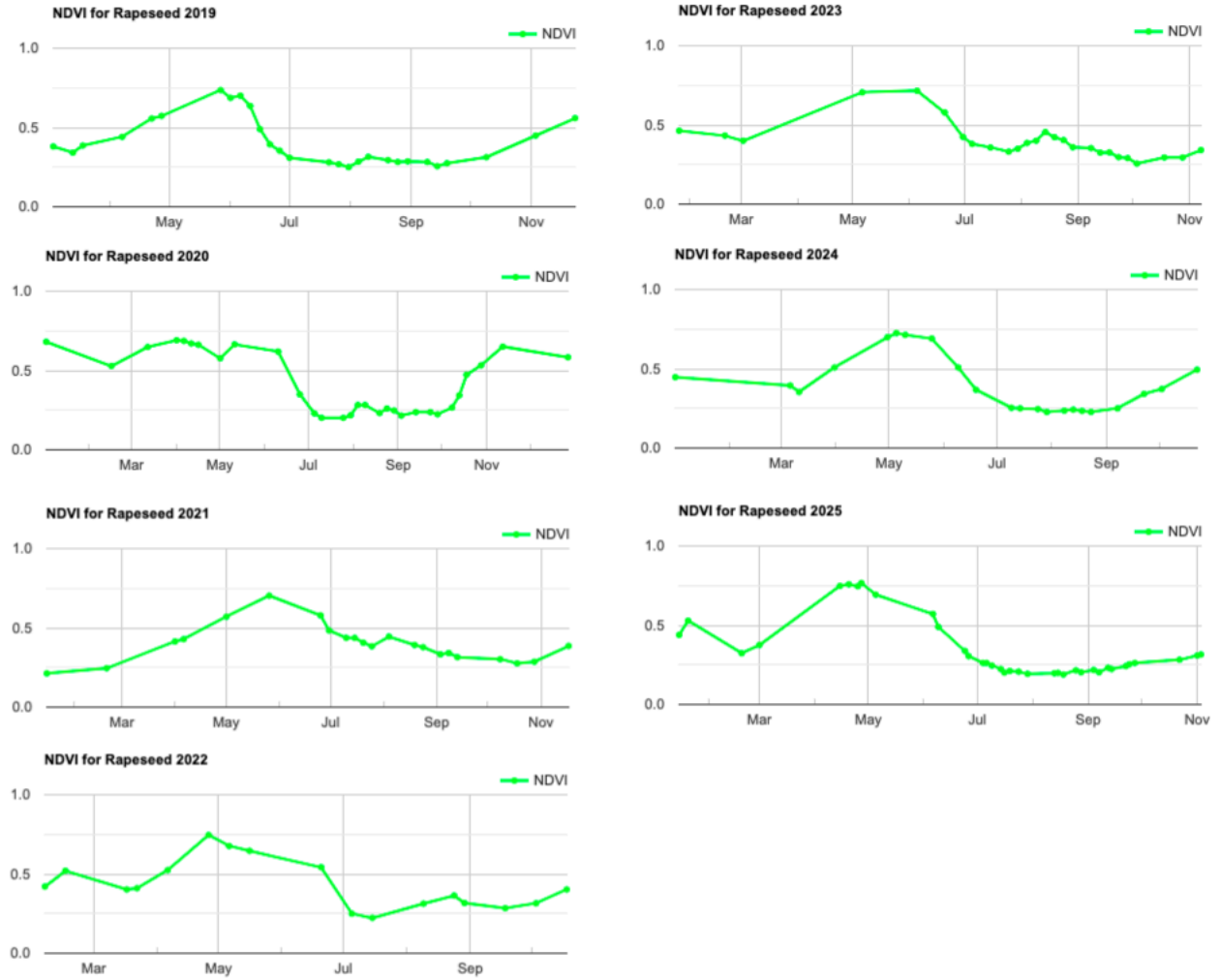


Figure A. NDVI Timeseries for 2019-2025 study years

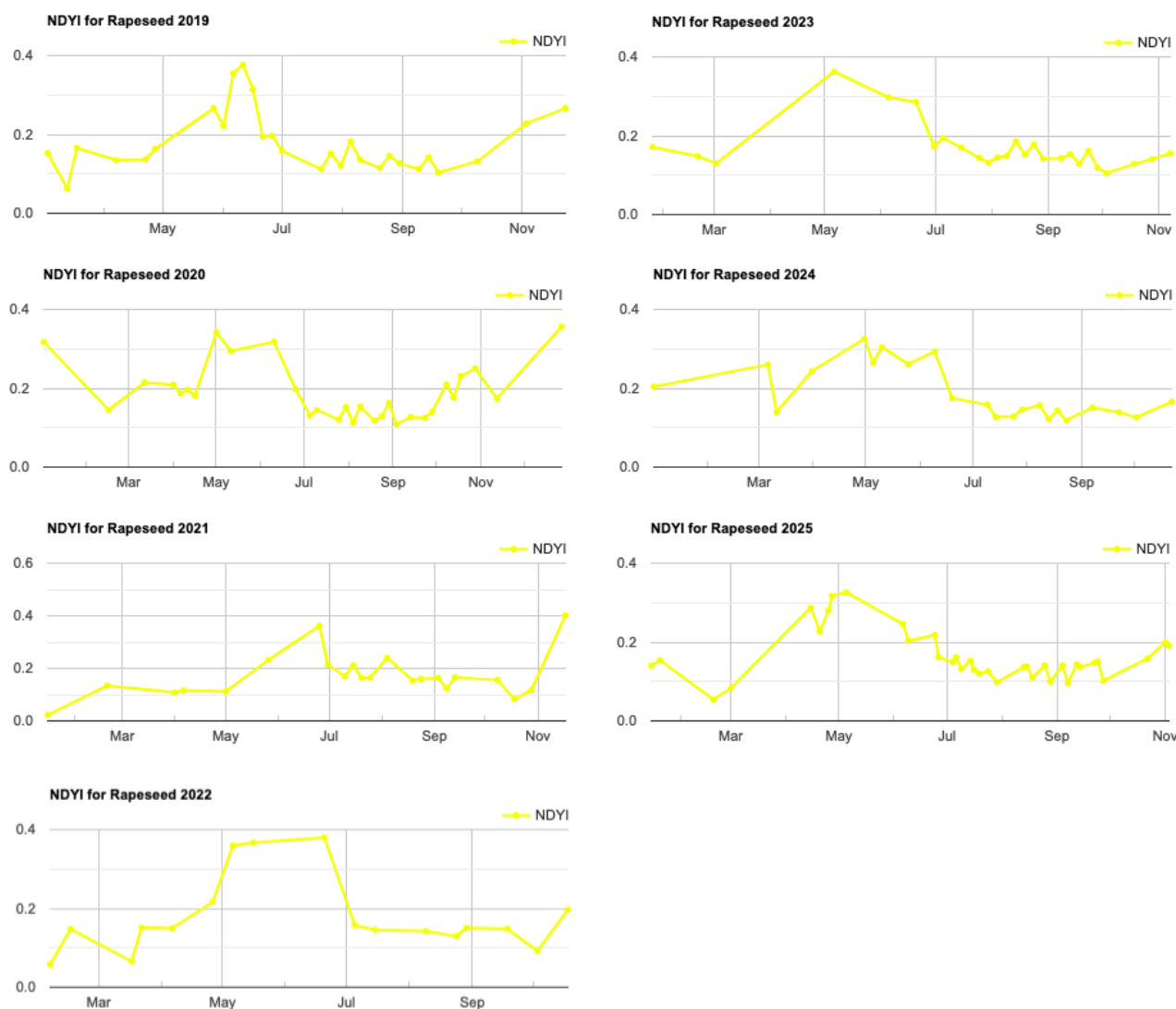
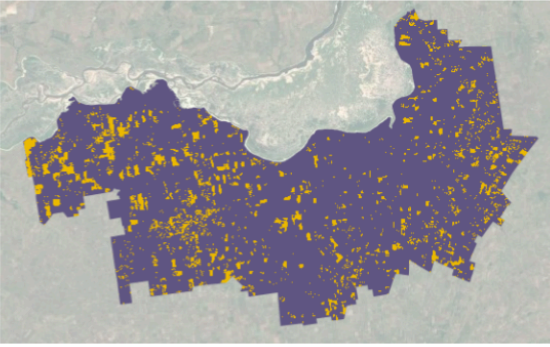
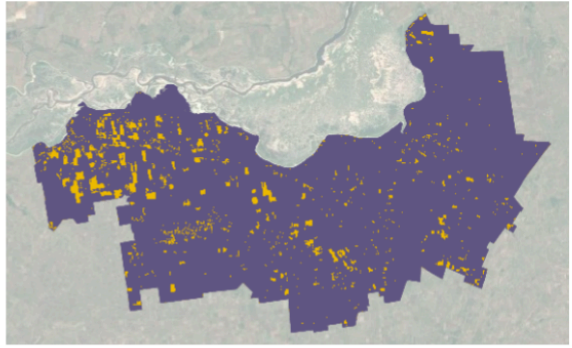


Figure B. NDYI Timeseries for 2019-2025 study years

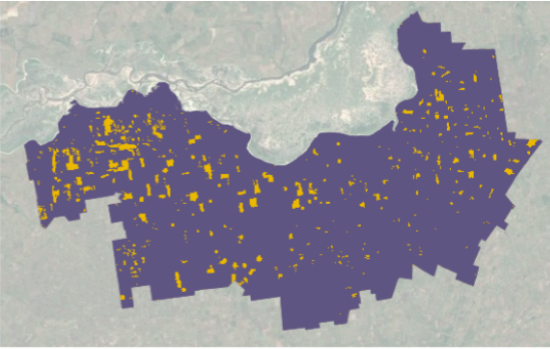
2019



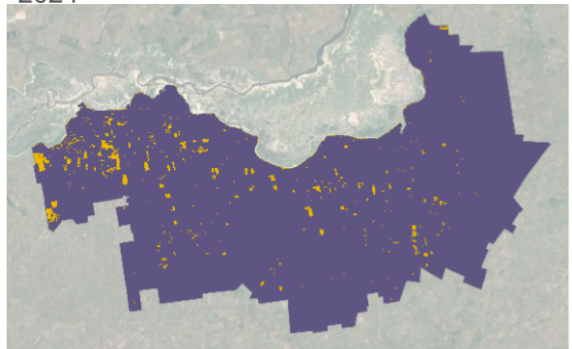
2023



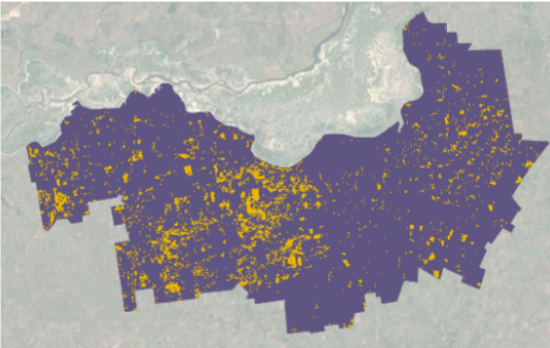
2020



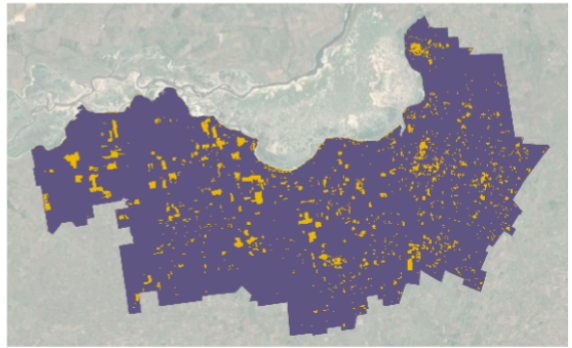
2024



2021



2025



2022

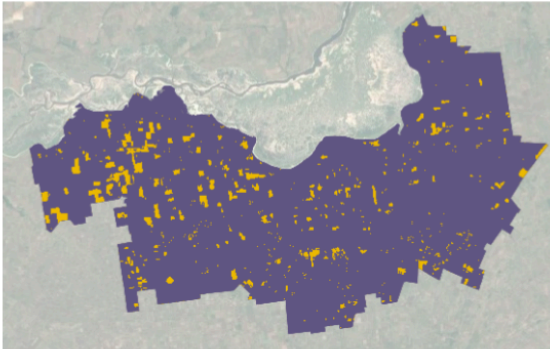


Figure C. Rapeseed classification maps for 2019-2025 study years.